

Cracking Query Bottlenecks: Towards Efficiency-Oriented Text-to-SQL Generation

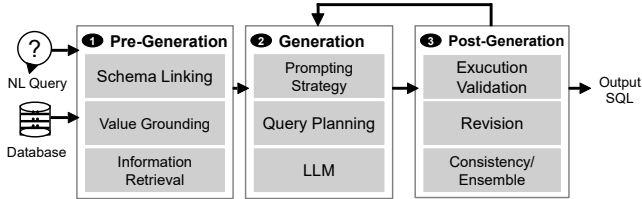


Fig. 1: Overview of a typical Text-to-SQL pipeline, illustrating the Pre-Generation, Generation, and Post-Generation stages

Abstract—Pending...

I. INTRODUCTION

Text-to-SQL aims to translate natural language (NL) questions into executable SQL queries, enabling non-expert users to retrieve and analyze relational data without manually writing database programs [1], [2], [3]. Its importance has grown with increasingly large schemas, rich database contents, and diverse analytical tasks, motivating benchmarks such as SPIDER [4], BIRD [5]. Recent LLM-based systems have achieved substantial progress through in-context learning, task decomposition and specialized agents, moving the field beyond single-shot generation toward workflow-based SQL construction [6], [7], [8], [9].

From Single-shot Generation to Agentic Workflows. As shown in Figure 1, modern Text-to-SQL systems typically organize SQL construction into three stages: pre-generation, generation, and post-generation. In the pre-generation stage, the system grounds the natural language query in the database context through schema linking, value grounding, and information retrieval, thereby reducing the search space and enriching the input evidence [9], [10], [11], [12]. In the generation stage, the model applies prompting strategies, query planning, and LLM-based reasoning to produce one or more candidate SQL queries [6], [7], [8], [13], [14], [15]. In the post-generation stage, these candidates are further checked and refined through execution validation, repair, and consistency- or ensemble-based selection [16], [7], [17], [8], [9], [13], [14], [15]. Compared with single-pass generation, such workflows are more robust because they decompose SQL construction into modular steps, strengthen database grounding, diversify candidate generation, and provide post-hoc opportunities for validation and refinement. Despite these advances, existing Text-to-SQL systems still face three coupled limitations:

L1: Limited Adaptation in Workflows and Strategy Selection. Many existing Text-to-SQL agents often predefine

both the workflow structure and the strategy choices, where modules such as schema linking, few-shot retrieval, question decomposition, candidate generation, validation, and revision are applied in a largely fixed manner [6], [7], [8], [13], [11]. Although such designs improve robustness over single-shot generation, they provide limited support for adapting reasoning strategies to the current database environment, question difficulty, or intermediate feedback. This can misallocate reasoning effort: simple queries over intuitive schemas may incur unnecessary token and latency overhead, while difficult queries may require deeper grounding, candidate comparison, or additional validation and reflection. This mismatch is illustrated by the different effects of domain-specific hints in BIRD [5]. For an intuitive database such as *superhero*, where tables and columns are relatively descriptive, adding domain hints provides little additional benefit in our experiments. In contrast, for a domain-heavy database such as *thrombosis_prediction*, where terms such as LDH and gender-dependent uric-acid ranges require medical knowledge, the same hints improve accuracy from 20% to nearly 50%. This contrast suggests that Text-to-SQL agents should adapt their grounding and generation strategies to the database environment, rather than applying the same strategy uniformly.

L2: Post-hoc rather than Process-level Reflection. Feedback in many Text-to-SQL systems is mainly used after a candidate SQL has been generated, such as repairing execution errors, revising invalid queries, or selecting among completed candidates [7], [8], [9], [18]. Such feedback improves final-output validation, but provides limited support for monitoring and regulating the generation process itself. In particular, the agent often lacks a controller that continuously tracks whether schema linking is reliable, whether candidate disagreement is increasing, whether repeated repairs are converging, or whether the current strategy should be changed. As a result, early mistakes in schema linking, value grounding, or query planning may propagate into later steps and become cascading hallucinations. Moreover, without global process supervision, local repair actions may fix surface-level execution errors while leaving the overall reasoning trajectory insufficiently grounded in the user question and database context.

L3: Limited Awareness of Query Efficiency. Most existing Text-to-SQL systems are primarily optimized for correctness: they aim to generate SQL queries whose execution results match the ground-truth answers [19], [20], [21], [22], [23], [24], [6], [7]. However, in practical database applications, a semantically correct SQL query is not necessarily efficient. Equivalent or near-equivalent SQL formulations may lead to

very different execution costs due to redundant projections, unnecessary joins, correlated subqueries, missing filter push-down, suboptimal join orders, or functions applied to indexed columns [25], [26], [27]. Although database systems have long relied on rule-based, cost-based, and learned rewriting to improve query efficiency [28], [27], [29], [30], current Text-to-SQL agents usually use database feedback mainly to check executability or result consistency. They rarely treat execution plans, estimated costs, or bottleneck patterns as signals for deciding whether and how to rewrite the generated SQL. As a result, the final query may be correct but unnecessarily expensive, especially in enterprise settings with large schemas, large tables, and long multi-step analytical workflows [18].

Key Insights: Meta-cognitive and Environment-aware Text-to-SQL. Inspired by enactive cognition and meta-cognition, we view Text-to-SQL not as a static translation from question and schema to SQL, but as an adaptive interaction process between the agent and the database environment [31], [32]. Enactive cognition suggests that understanding is shaped through interaction with the environment, while meta-cognition emphasizes the ability to monitor and regulate one’s own reasoning process. In the context of Text-to-SQL, this means that an agent should perceive database-specific signals, act by generating and rewriting SQL, observe feedback from execution and query plans, and adjust its strategy when uncertainty, failure, or inefficiency is detected. These perspectives lead to three design objectives. First, to address **L1**, the agent should perform *environment-aware workflow adaptation via strategy selection*: it should select lightweight or deep workflows according to the perceived database environment, question complexity, and intermediate feedback, rather than applying the same strategy uniformly. Second, to address **L2**, the agent should support *process-level self-reflection and dynamic control*: it should monitor schema-linking reliability, candidate disagreement, repair convergence, and strategy suitability during generation, so that early mistakes can be corrected before they propagate. Third, to address **L3**, the agent should enable *execution- and plan-aware SQL optimization*: it should use query plans, cost signals, semantic validation, and reusable rewrite knowledge to improve query efficiency while preserving correctness.

II. MOTIVATION

A. Task Formulation

Traditional Text-to-SQL aims to translate a NL question x into an executable SQL query q over a database environment $\mathcal{D}$. Given a ground-truth query q^* , the standard objective is to generate a query whose execution result matches that of q^* :

$$f_\theta(x, \mathcal{D}) \rightarrow q, \quad \text{s.t.} \quad \text{Exec}(q, \mathcal{D}) = \text{Exec}(q^*, \mathcal{D}),$$

where f_θ denotes a Text-to-SQL model parameterized by θ , $\mathcal{D}$ denotes the database environment including schema, data, indexes, and statistics, and $\text{Exec}(q, \mathcal{D})$ denotes the result of executing q on $\mathcal{D}$.

However, this formulation focuses mainly on semantic correctness and does not explicitly consider query efficiency.

In practice, multiple semantically equivalent SQL queries may produce the same result but incur substantially different execution costs due to join order, predicate placement, aggregation strategy, or index utilization. We therefore formulate **Efficiency-Oriented Text-to-SQL**, where the goal is to generate a query that is both semantically correct and efficient. Given a NL question x and a database environment $\mathcal{D}$, the goal is to generate a SQL query q_{opt} satisfying:

$$q_{\text{opt}} = \arg \min_{q \in \mathcal{Q}_{\text{corr}}(x, \mathcal{D})} \text{Cost}(q, \mathcal{D}),$$

$$\mathcal{Q}_{\text{corr}}(x, \mathcal{D}) = \{q \in \mathcal{Q}(x, \mathcal{D}) \mid \text{Exec}(q, \mathcal{D}) = \text{Exec}(q^*, \mathcal{D})\},$$

where $\mathcal{Q}_{\text{corr}}(x, \mathcal{D})$ denotes the subset of candidates that are semantically correct with respect to q^* . $\text{Cost}(q, \mathcal{S})$ represents the estimated or actual execution cost of query q on $\mathcal{S}$. Intuitively, the model should not only ensure that q_{opt} yields correct results but also that it achieves minimal execution time or resource consumption under the same semantic constraints.

B. Design Objective

Existing Text-to-SQL systems are typically designed as static pipelines that decompose query generation into pre-defined stages, including schema linking, candidate generation, and post-hoc execution validation. While such designs improve robustness compared to single-pass generation, they lack the ability to dynamically adapt their reasoning process to different database environments, query difficulties, and intermediate execution feedback.

Beyond Static Pipelines: A Meta-Cognitive Multi-Agent Perspective. Inspired by enactive cognition and meta-cognitive theory, we reformulate Text-to-SQL as a dynamic interaction process between an intelligent system and the database environment. Enactive cognition emphasizes that intelligence emerges through continuous interaction with the environment, while meta-cognition enables the system to monitor, regulate, and adapt its own reasoning process. Based on these principles, we propose a Meta-Cognitive Multi-Agent Text-to-SQL System, where a central Meta-Cognitive Controller dynamically orchestrates a set of specialized agents (actions) under different execution contexts. Instead of a fixed pipeline, all stages in Text-to-SQL are decomposed into reusable and composable actions, which are invoked online based on the current state of the problem and environment feedback. This design establishes a unified closed-loop system in which a Meta-Cognitive Controller coordinates planning, execution, evaluation, and optimization over SQL programs based on continuous database feedback.

Key Design Principles. (1) Environment-aware Workflow Adaptation via Strategy Selection. The Meta-Cognitive Controller explicitly assesses both the database environment and the complexity of the input query to determine the appropriate reasoning strategy. For example, for simple queries over well-structured schemas (e.g., single-table filtering), the controller may directly invoke a lightweight generation path using few-shot prompting. In contrast, for complex analytical queries (e.g., multi-join or domain-specific reasoning), the controller

activates a deeper workflow that includes multi-candidate generation, iterative refinement, and optimization loops. This adaptive strategy selection avoids over- or under-utilization of reasoning resources.

(2) Process-level Self-Reflection and Dynamic Control.

Unlike conventional post-hoc validation approaches, our system enables continuous monitoring of intermediate reasoning states during SQL generation. The Meta-Cognitive Controller observes signals such as schema linking ambiguity, candidate disagreement across generated SQLs, and convergence behavior in optimization loops. For instance, if multiple candidate SQLs consistently disagree on join paths, the controller may trigger re-linking of schema entities or restart generation with alternative evidence. Similarly, if repeated optimization iterations fail to reduce execution cost, the controller may terminate the current loop and switch to a different reasoning configuration. This enables closed-loop control over the entire reasoning trajectory.

(3) Execution- and Plan-aware Optimization via Feedback-driven Rewriting.

The system treats execution results, query plans, and cost estimations as fundamental feedback signals that continuously shape its optimization behavior. When execution traces reveal inefficiencies such as full table scans, suboptimal join orders, or redundant computation paths, the system invokes an optimization agent that reasons over both the current execution feedback and previously observed optimization experiences to produce improved SQL variants. Crucially, optimization is not performed in isolation for each query. Instead, the system continuously accumulates experience from execution outcomes and rewriting actions, forming an evolving Auto-Evaluation Knowledge Base (AEKB). This knowledge base stores historical rewrite cases, execution feedback, and observed performance patterns, enabling the agent to generalize from past optimization experiences to new queries. In this way, query optimization is transformed from a rule-based or purely local rewriting process into an experience-driven learning loop, where execution feedback serves as the foundation for progressively improving future reasoning and optimization decisions.

III. OPTSQL

A. Overview

Figure 2 shows the overall architecture of OPTSQL. Given a NL question and a database environment, OPTSQL aims to generate a SQL query that is both semantically correct and execution-efficient. Instead of treating Text-to-SQL as a fixed pipeline, OPTSQL formulates it as a feedback-driven process coordinated by a Meta-Cognitive Controller. The Meta-Cognitive Controller supervises the entire workflow. It perceives the database environment, estimates the difficulty of the input question, selects suitable actions, and monitors intermediate feedback throughout both generation and optimization. Under its control, OPTSQL first constructs an initial SQL query in the Generation Phase, where an evidence-guided schema filter identifies relevant tables, columns, values, and

join paths, and an initial SQL builder produces executable candidate queries. If the controller detects that the query requires further efficiency improvement, it activates the Optimization Loop. This loop consists of an Explain Analyzer Agent, a SQL Rewriter Agent, and a Validator Agent, which together analyze execution plans, rewrite inefficient SQL patterns, and verify semantic consistency and cost reduction. OPTSQL is further supported by a Memory Engine and a Tool Infrastructure. The Memory Engine maintains both task-level working memory and reusable optimization knowledge, including expert rules, historical rewrite cases, and observed optimization patterns. The Tool Infrastructure provides database grounding, SQL execution and analysis, cost evaluation, semantic consistency checking, and RAG-based retrieval. Through these components, OPTSQL continuously observes feedback from the database environment, reflects on its intermediate decisions, and updates its optimization experience.

The remainder of this section describes each component of OPTSQL. We first present the Generation Phase for constructing an initial SQL query (Section III-B), followed by the Optimization Loop for execution-aware rewriting (Section III-C). We then introduce the Meta-Cognitive Controller that orchestrates these processes through strategy selection, monitoring, and self-reflection (Section III-D). Finally, we describe the Memory Engine and Tool Infrastructure that support retrieval, experience accumulation, and database feedback (Section III-E).

B. Generation Phase

The Generation Phase aims to construct a set of executable and semantically plausible initial SQL candidates based on the question and database environment. It focuses on two core capabilities: evidence construction and multi-strategy SQL generation, without global control or optimization decisions.

Evidence-guided Schema Filter Agent. This agent is responsible for grounding the input question into a compact, relevant, and executable schema context for downstream SQL generation. This agent is responsible for grounding the input question into a compact, relevant, and executable schema context for downstream SQL generation. It combines robust schema linking with value-level grounding, aiming to identify not only the tables and columns mentioned or implied by the question, but also the concrete database values and conditions needed to express the user intent. To improve schema robustness, the agent supports multiple complementary schema linking strategies:

- *Direct schema linking* maps the natural language question to relevant tables and columns by reasoning over question semantics and schema descriptions. This strategy is effective when the question contains explicit lexical or semantic cues.
- *Reverse schema linking* hypothesizes a tentative SQL structure and then extracts the tables and columns implied by this structure [33]. This recovers latent schema dependencies, such as join tables, grouping columns, or aggregation-related attributes, that may be missed by direct matching.

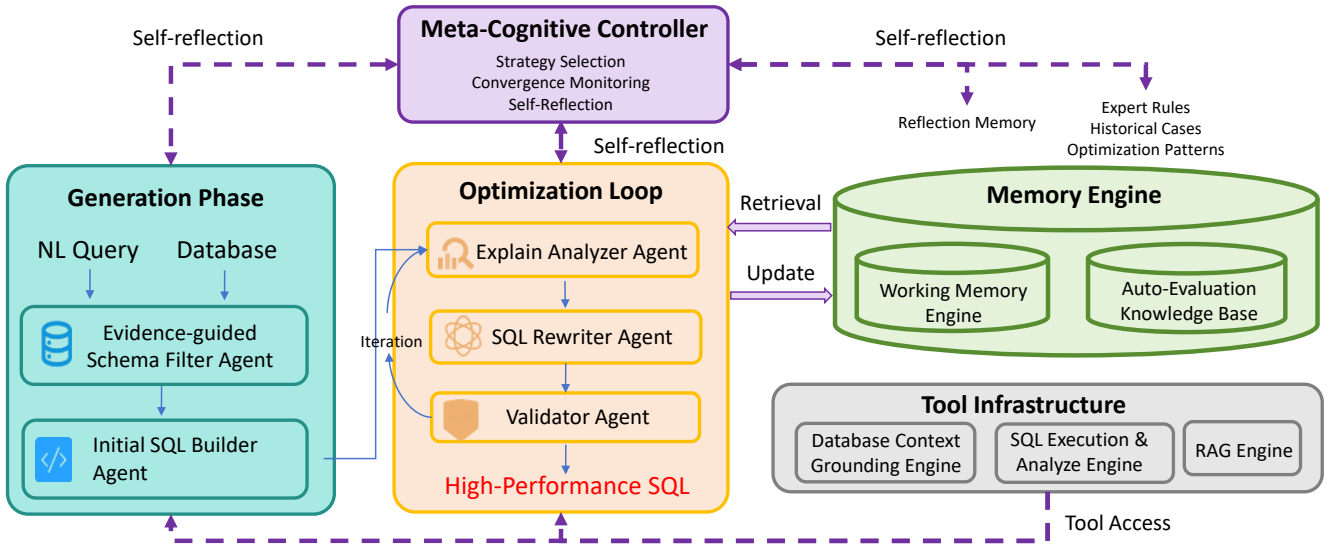


Fig. 2: Overview of OPTSQL

- *Value-grounded schema linking* identifies relevant schema elements through actual database contents. Instead of relying only on table or column names, the agent checks whether values in a column match or semantically correspond to phrases in the question, which is especially useful when schema names are ambiguous or weakly informative.

Beyond schema linking, the agent further performs active value grounding to translate natural language expressions into concrete database predicates. It supports several value grounding capabilities:

- *Exact value verification* checks whether a phrase in the question directly appears in candidate columns. This is useful for grounding categorical conditions such as names, locations, statuses, or labels.
- *Semantic value matching* retrieves column values that are semantically close to question phrases, allowing the agent to handle paraphrases, abbreviations, and domain-specific expressions.
- *Distinct-value inspection* examines representative or unique values of a column to infer how natural language concepts are encoded in the database, such as whether a Boolean concept is represented by 0/1, Y/N, or textual labels.
- *Data sampling and distribution checking* inspects example rows or value distributions to understand column semantics and avoid grounding predicates on irrelevant or sparsely populated columns.
- *Null and type analysis* checks whether candidate columns contain nulls or unexpected value formats, helping the agent avoid unreliable predicates and distinguish categorical, numerical, date, and Boolean constraints.
- *Composite predicate grounding* combines multiple grounded values or columns when a natural language phrase corresponds to a compound condition, such as a grade span, date range, or multi-attribute status.

For example, a phrase such as “kindergarten to 8th grade” may be grounded as `Low_Grade = 'K'` and `High_Grade`

`= '8'`, while “magnet program” may correspond to a binary or categorical value after inspecting the value distribution of a related column. These grounded facts are passed to the SQL generator as explicit evidence, reducing the risk of hallucinated predicates or incorrect value choices. Finally, the agent further completes the relational context required for SQL construction. The selected schema elements may be semantically relevant but structurally disconnected; for example, two relevant tables may require intermediate tables or primary–foreign key columns to form a valid join path. Therefore, the agent supplements necessary join keys and bridge tables according to the database relationships, so that the resulting schema context can support executable SQL generation. The output of this agent is an evidence-enriched schema context that integrates schema relevance, verified value grounding, ambiguous-case evidence, and join-path information for downstream SQL generation.

Initial SQL Builder Agent. Initial SQL Builder Agent is responsible for producing executable and semantically plausible initial SQL candidates. It provides a set of complementary SQL construction capabilities, rather than a fixed generation pipeline. Depending on the task complexity, schema ambiguity, and control signals from the Meta-Cognitive Controller, the agent may invoke one or more generation strategies to balance robustness and efficiency. The agent supports the following SQL generation strategies:

- *Skeleton-based generation* follows a top-down construction process. It first analyzes the required SQL components, such as selection, filtering, joining, grouping and ordering, then sketches a SQL skeleton, and finally fills in concrete tables, columns, predicates, and functions. This strategy is useful for queries that require clear structural planning.
- *ICL-based generation* leverages retrieved demonstrations to guide SQL construction. By retrieving structurally similar examples, the agent can reuse successful query patterns and adapt them to the current grounded schema. This strategy is especially helpful when the question resembles previously

observed SQL structures.

- *Decomposition-based generation* breaks a complex question into simpler sub-questions and generates SQL fragments for each sub-task before composing them into a complete query. This strategy is designed for complex analytical questions involving nested reasoning, multiple conditions, or multi-step aggregation.

After candidate generation, the agent can apply a lightweight repair loop to improve executability. It checks basic syntactic and execution errors, and revises invalid candidates using the original question, grounded schema evidence, and database error messages. This repair process removes obvious generation failures before candidate selection, while leaving global strategy switching and deeper optimization to the Meta-Cognitive Controller. When multiple candidates are available, the agent further supports *execution-based selection*. Candidates are grouped according to their execution results, and each group is treated as a behavioral cluster. If one cluster forms a dominant majority, the agent selects a representative query from that cluster as the initial SQL. Otherwise, representative candidates from the top clusters are compared through pairwise semantic judgment, and each candidate is ranked by a confidence-aware score. The candidate with the highest score is selected as the output of the Initial SQL Builder Agent.

C. Optimization Loop

The Optimization Loop aims to improve the execution efficiency of an initial SQL query while preserving its semantics. It is activated when the Meta-Cognitive Controller determines that the generated SQL may benefit from further optimization, such as when the query involves large tables, complex joins, or inefficient query plans. Instead of treating SQL generation as the final step, this loop refines the initial query through execution-aware analysis, rewriting, and validation.

D. Meta-Cognitive Controller

E. Memory Engine and Tool Infrastructure

IV. EXPERIMENT

V. DISCUSSION

VI. RELATED WORK

Fine-tuned and Trained Text-to-SQL Models. A dominant line of recent work focuses on adapting or training LLMs for Text-to-SQL via supervised fine-tuning, synthetic data construction, or reinforcement learning [34], [20], [35], [36], [37], [38], [39], [40]. For example, X1YAN-SQL [39] uses multi-task fine-tuning and a multi-generator ensemble to produce diverse SQL candidates, while OMNISQL [40] improves model training through large-scale synthesized Text-to-SQL data. Reinforcement-learning-based methods such as REASONING-SQL [41] and ARCTIC-TEXT2SQL-R1 [42] further train models with SQL-tailored rewards or execution-based feedback to improve reasoning and executability. Despite their strong performance on benchmarks, these methods remain heavily dependent on training data quality and coverage. They often exhibit limited generalization in out-of-domain

scenarios and may suffer from reduced robustness when facing complex or real-world database environments [33].

LLM Prompting and Retrieval-augmented Generation. Another line of work leverages general-purpose LLMs for Text-to-SQL through prompting and retrieval-augmented generation, performing task adaptation at inference time without parameter updates. These methods formulate SQL generation as an in-context reasoning problem, where the model is guided by schema serialization, database content, demonstrations and decomposition instructions. Early benchmark and adaptation studies show that LLMs can become strong Text-to-SQL generators when prompts provide sufficient database context and carefully selected examples [6], [5], [43]. Subsequent prompting methods improve robustness through task decomposition, self-correction, dynamic few-shot selection, and consistency alignment [7], [44], [45]. Retrieval-augmented approaches further strengthen grounding by retrieving relevant demonstrations, linking schema elements at scale, pruning database context, and selecting among multiple candidates [10], [9], [46], [47], [48]. Despite these advances, prompting and retrieval-based methods still depend heavily on manually designed inference procedures and the quality of retrieved evidence. Their reasoning depth is usually determined before generation, which limits their ability to adapt the workflow according to schema scale, intermediate uncertainty, or execution behavior [33], [9].

Agentic Text-to-SQL Workflows. Recent work further extends LLM prompting into agentic workflows, where LLMs call tools, inspect database context, execute SQL, observe feedback, and revise intermediate outputs. General agent frameworks such as REACT [17], SELF-REFINE [49], and REFLEXION [50] demonstrate the value of combining reasoning, acting, and iterative self-improvement. Following this paradigm, Text-to-SQL agents organize generation around modules such as schema exploration, candidate generation, execution-based repair, multi-path reasoning, semantic memory, and agentic view construction [8], [46], [51], [13], [14], [15], [11], [12]. These systems are more robust than single-shot prompting because they can query the database, compare alternatives, and use observed failures to revise SQL. Nevertheless, most existing agents still instantiate a largely fixed workflow: similar modules are invoked for most questions, and database feedback is mainly used for post-generation repair, validation, or candidate selection rather than for continuous process control.

Difference from OPTSQL. Existing prompting-based and agentic Text-to-SQL methods mainly focus on improving correctness through better grounding, decomposition, and post-hoc refinement. In contrast, OPTSQL is designed as an efficiency- and process-aware framework that goes beyond static or fixed workflows. It introduces a meta-cognitive controller that enables (i) environment-aware workflow adaptation, (ii) process-level self-reflection during generation, and (iii) execution- and plan-aware optimization guided by database feedback. As a result, OPTSQL treats Text-to-SQL as a dynamic interaction process where both reasoning strategy and

optimization behavior are continuously adjusted according to the database environment and intermediate execution signals, rather than following a predefined pipeline or relying only on post-generation correction.

VII. CONCLUSION

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